MIL-R-60399 (MU)
1 November 1966

MILITARY SPECIFICATION

REDUCER, FLASH, M1, FOR 155MM GUN

1. SCOPE

1.1 This document covers the loading, assembling and packing for one type of flash reducer designated as M1 for use in 155MM Gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-P-291 - Paperboard, Wrapping, Cushioning
PPP-T-60 - Tape, Pressure-Sensitive Adhesive, Waterproof, For Packaging

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-T-43036 - Tape, Pressure-Sensitive Adhesive, Plastic Film, Fildment Reinforced (For Sealing Fiber Containers and Cans)
MIL-B-43165 - Bag, Polyethylene
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

FSC 1320

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

76-1-1039 - Drum, Metal, For 75MM and 105MM Howitzer
9217087 - Marking Diagram and Sealing of Steel Drums for Shipment of Propellants and Bagged Charges.
9218468 - M1 Flash Reducer

PUBLICATIONS

ARMY

ET-9218468 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material shall be in accordance with applicable drawings and specifications.

3.2 Flash reducer assembly.-The flash reducer assembly shall comply with all requirements specified on Drawing (dwg.) 9218468 and with all requirements specified in applicable specifications.

3.3 Workmanship.-All charges shall be free of dirt, grease and other foreign material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Each lot shall contain:

a. Potassium sulfate from not more than one lot from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified,) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

4.2.2.1 Body (see dwg. 9218469, covered by dwg. 9218472 covering a detail of dwg. 9218468).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 percent		
201.	Length	Gage	01001
202.	Width	Gage	01002
203.	Cloth improper including incorrect color	Visual	01003
204.	Marking missing, misleading or unidentifiable	Visual	01004
205.	Evidence of poor workmanship (see 3.3)	Visual	01005

4.2.2.2 Tying tape (see dwg. 9218474 covering by dwg. 9218470 9218471, 9218472, covering a detail of dwg. 9218468).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
102.	Length	Gage	02001
Minor:	AQL 1.5 percent		
201.	Width	Gage	02002
202.	Cloth improper, including color	Visual	02003
203.	Evidence of poor workmanship (see 3.3)	Visual	02004

4.2.2.3 Increment charge assembly (see dwg. 9218470 covering a detail of dwg. 9218468).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

Minor:		AQL 2.5 percent		
201.	Distance from tape to body ...		Gage	03001
202.	Distance between bodies		Gage	03002
203.	Distance from body end to tape		Gage	03003
204.	Depth of pocket		Gage	03004
205.	Any seam or opening incompletely stitched		Visual	03005
206.	Evidence of poor workmanship (see 3.3)		Visual	03006

4.2.2.4 Base charge upper assembly (see dwg. 9218471 covering a detail of dwg. 9218468).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.5 percent		
201.	Distance from tape to body	Gage	04001
202.	Distance between bodies	Gage	04002
203.	Distance from body end to tape	Gage	04003
204.	Depth of pocket	Gage	04004
205.	Any seam or opening incompletely stitched	Visual	04005
206.	Marking missing, misleading or unidentifiable	Visual	04006
207.	Evidence of poor workmanship (see 3.3)	Visual	04007

4.2.2.5 Base charge lower assembly (see dwg. 9218472 covering a detail of dwg. 9218468)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

Minor:	AQL 2.5 percent		
201.	Distance from tape to body ..	Gage	05001
202.	Distance between bodies	Gage	05002
203.	Distance from body end to tape	Gage	05003
204.	Depth of pocket	Gage	05004
205.	Any seam or opening incompletely stitched	Visual	05005
206.	Evidence of poor workmanship (see 3.3)	Visual	05006

4.2.2.6 Flash reducer (see dwg. 9218468).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Assembly damaged to the extent that charge or black powder can escape	Visual	06001
102.	Charge weight out of tolerance	Balance	06002
103.	Black powder weight out of tolerance	Balance	06003
Minor:	AQL 1.5 percent		
201.	Any seam or opening incompletely stitched	Visual	06004
202.	Thread broken	Visual	06005
203.	Evidence of poor workmanship (see 3.3)	Visual	06006

4.2.2.7 Drum, prior to closing (see 5.2)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Polyethylene bag missing	Visual	07001
102.	Polyethylene bag improperly sealed	Visual	07002
Minor:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

4.2.2.8 Drum, sealed (see 5.2 and 5.3)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Lid loose or improperly assembled	Manual	08001
Minor:	AQL 1.0 percent		
201.	Contents move when shaken ...	Manual	08002
202.	Marking missing or misleading or unidentifiable	Visual	08003

4.2.3 Inspection Equipment.-Equipment Tabulation Number ET-9218468 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.3.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.3.2 Supply and maintenance.-Supply and maintenance of the equipment listed in the ET shall be in accordance with Specification MIL-I-45607.

4.2.3.3 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A.-Each flash reducer shall be folded to a package approximately 12 by 4 by 1-3/4 inches thick.

5.2 Packing

5.2.1 Level A.-Fifty packages shall be packed in a metal drum conforming to drawing 76-1-1039. The interior of the drum shall be completely lined with one or more layers of paperboard cushioning material conforming to PPP-P-291, Type III, Style I. Additional paperboard shall be added as required to provide a tight pack. The lid shall be securely tightened on the drum.

5.3 Marking.-Each container shall be marked in accordance with drawing 9217087. The ICC nomenclature shall be, "BLACK POWDER".

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspectio.. of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NDl. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1320-A115

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T00:35Z
Check the source to verify that this is the currenversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

5. SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

COMMANDING OFFICER
United States Army
Picatinny Arsenal
ATTN: SMUPA-ND
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:35Z
Check the source to verify that this is the current version before use.